

# Program Manager | Digital & Scientific Transformer | Project Director | AI Research Fellow | Mentor-21st Century STEM Skills Kudakwashe Dandajena

Ph.D. (Candidate) Computer Science (Artificial Intelligence)  
MSc Computer Science,  
BSc (Hons) Computer Science,  
Cert Project Management (Prince2),  
Cert High-Performance Computing Engineering and  
Cert Data Centre Engineering

## About Me

I am a motivated, innovative, and seasoned specialist with 11+ years of global experience in managing non-governmental/non-profit organizations, the public sector, digital transformation, infrastructure, scientific research, and academic programs and projects (*cumulative value exceeding US\$100 million*) in agile environments. Excels in the formation of sustainable partnerships, collaborations, product co-creation, consulting and advisory services, and fundraising.

## Summary of Achievements

- 2022-Present; Mobilized funding for implementing the ongoing Regional Blended Next Einstein Forum Africa Science Week 2022 Edition STEM program across five African Regions (Northern—Tunisia, Western—Ghana, Southern—South Africa, Eastern—Rwanda, and Central—DRC) as **Program Manager**.
- 2022-Present; Participated in the development of the **10 year Mastercard Foundation Scholars Program Partners** funding program for the African Institute for Mathematical Sciences (AIMS) Global Network as a **Team Member**.
- 2021-Present; Established 10 strategic partnerships across the African Institute for Mathematical Sciences (AIMS) global network (Africa, North America, and Europe) within a period of nine months with global development agencies, industry, and scientific institutions for STEM-related programs towards the transformation of Africa as **Program Manager**.
- 2020-21; provide technical advisory services to the Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)-FAIR Forward Artificial Intelligence for AI Program in Rwanda as an Artificial Intelligence and Machine Learning **Lead Specialist** for the African Leadership University.
- 2020; Transformed the learning model at African Leadership University in Rwanda by introducing a hybrid industrial driven mastery learning model as an Artificial Intelligence and Mathematics **Facilitator and Course Director**.
- 2020; Developed and implemented the National Digital Transformation Agenda Phase 1 for the Government of Zimbabwe. This enabled seamless digital government business during the Covid-19 pandemic period as a **Program Manager**. Some major project responsibilities were strategy formulation; management; execution, monitoring and budgeting (CAPEX and OPEX) of the whole digital journey.
- 2019; Implemented the first National ICT Manufacturing Plant in Zimbabwe called ZiTCO, for the manufacturing and assembly of ICT devices such as tablets, computers, electricity and water smart meters and other ICT gadgets. Played a conceiving or initiation, planning, coordinating, financial mobilization, procurement, implementation and operationalization role as the **Project Manager**.

- 2019; Implemented the Zimbabwe National Data Centre Project for Public Cloud Computing Services as part of the national digital transformation agenda as the **Project Manager**. Responsible for project funding, planning, implementation and coordination.
- 2019; Executed the Government Communication Modernization Project on scope, time and cost which transformed the nationwide communication systems from siloed, expensive and manual environment through modern technology platforms. The project was implemented in collaboration with World Bank and IMF. Responsible for all project implementation matter as a **Digital Transformer and Project Manager**.
- 2018; Implemented the Zimbabwe National Critical Skills Audit Project of 2018 as the **Project Coordinator** in collaboration with 'the African Capacity Building Foundation (ACBF) and the World Bank under the Ministry of Higher and Tertiary Education Science and Technology Development.
- 2015 - 2017; Implemented the National Science Technology Engineering and Mathematics (STEM) Programme in 2016 as the **Lead Project Coordinator**.
- 2013 - 2014; Pioneered and implemented the first Supercomputing or High-Performance Computing (HPC) facility in Zimbabwe called the Zimbabwe Centre for High Performance Computing (ZCHPC) as the **Project Manager**. Played a coordinating role in conceiving, fundraising, international bilateral negotiations, technical and engineering designing.
- 2012 - Present; Ongoing Established strategic technological supply chain links with top technology vendors, equipment manufacturers, distributors (OEMs and ODMs), system integrators (SIs) and solution providers as a **Senior Technology Expert**.
- 2011 - 2013; Designed and formulated National Cybersecurity Legal Framework (Cybercrime Act, Data Protection Act and Electronic Transactions and e-Commerce Act) Legislation in Zimbabwe in conjunction with International Telecommunication Union (ITU) and Southern African Development Community (SADC) as a **Technology Expert**.

#### Education

2019 - Present; Ph.D. Candidate (soon to be awarded)) in Computer Science Specializing in Deep Learning, University of the Western Cape

2015 - 2017, Nov; MSc Computer Science: National University of Science and Technology, Zimbabwe

2005 – 2009, Nov; BSc (Hon) Computer Science: Midlands State University, Zimbabwe

#### Professional Certifications and Special Training

2017; CompTIA+ Storage Training-Adept Solutions, Zimbabwe.

2017; Red Hat Linux (RHEL) Administration Training-Adept Solutions, Zimbabwe

2016; Supercomputing & Data Centre Engineering Cert- Inspur Group of China

2016; Project Management Certification (PRINCE 2)-SNIA

2015; Specialised High-Performance Computing Training-TACC, United States of America

2015; Cloud Computing and Big Data Cert -MOFCOM, China

2013; Supercomputing & Data Centre Engineering Cert-Inspur Group of China

2011; Advanced Scientific Data Centre Applications-Meteo France Int. Toulouse, France

#### Technological Platforms and Languages

- *Artificial Intelligence, Machine Learning and Deep Learning Platforms* — Microsoft Azure, Google AI and Colab, Kaggle, Python Libraries (Keras, Pandas, TensorFlow, Anaconda, Jupyter, Django, Flask etc.).
- *Data Science and Big Data Platforms* — Hadoop, Microsoft Power BI.
- *High Performance Computing* — CUDA, MPI, InfiniBand Communication, Hybrid Parallel Computing (GPU, CPU and TPU).
- *Operating Systems Platforms* — Linux (RHEL, Ubuntu and Centos), Windows, MacOS and Android
- *Collaboration and Project Management Platform* — Trello, Canvas, Slack, Microsoft Office Packages (MS Projects, Word etc), Virtual platforms (Zoom, MS Teams, GoogleMeet, etc.)
- *Programming Languages and Databases* — Python, Java, SQL, C and C++.

### Membership to Professional Organizations

2021– Current; Ex—Folio Board Member of the Executive Committee of the Next Einstein Forum Community of Scientists

2019 – 2020; Vice Chair of the Board of Management for Courtney Selous School, Zimbabwe

2017 – 2020; Board Member, Institute of Information Technology Professional, South Africa

2017 – 2019; Board Member, Australia Computer Society Membership Australia– Pending renewal.

### Employment History

Aug 2021 – Present; **Program Manager for the Community of Scientists** – African Institute for Mathematical Sciences (AIMS), Global Secretariat, Kigali, Rwanda: **NON—GOVERNMENTAL/PROFIT ORGANISATION**

- Responsible for managing Next Einstein Forum (NEF) Community of Scientists with over 10,000 members composed of NEF Fellows, NEF Ambassadors in 54 African Countries, AIMS alumnus, African science entrepreneurs (Science-preneurs), Researchers, Scientists, Inventors and Learners
- Management of partnerships, collaboration and fundraising with the Industry Partners, Public Sector, Non-Governmental/Profit Organisation, International Agencies, and Individuals
- Managing scientific programs and projects of the entire AIMS Global Network in Africa, Canada, United Kingdom, , Germany and Scandinavia
- In charge of technical program development for global and continental science initiatives and gathering such as Global Gathering (GG), Africa Science Week (ASW), Scientific African Journal (SAJ)).
- Managing entrepreneurship and innovation portfolio for scientific programs and projects
- Responsible for program advisory, support, research, development and innovation support.

July 2020 – Aug 2021; **Senior Computer Science Facilitator** – Faculty of Computer Science, African Leadership University, Kigali, Rwanda: **ACADEMIC INSTITUTION**

- Responsible for transformational leadership and mentoring ethical leaders, entrepreneurs, and problem solvers within the fourth industrial revolution space of Artificial Intelligence, Machine Learning, Data Science and Discrete Mathematics.
- Coaching, motivating, tracking and reviewing learner's technology designs, algorithms, and codes in line with global challenges and sustainable development goals (SDG's).
- Drove scientific research, development and innovation mission to over 300 ambitious learners and problem solvers with a mission to address existing global challenges within the context of sustainable development goals.
- Responsible for developing mathematical data science, machine learning, and artificial intelligence capabilities using industry driven high-performance architectures, algorithms, prototypes, models, visualization and proof of concepts.
- Demonstrating the power of next generation technologies and optimized processes to learners and its impact in developing and deploying industrial integrated digital enterprise solutions for future African techno-preneurs, leaders and problem solvers.
- Responsible for providing advisory solutions to the University, Multinational Partners and Government of Rwanda on digital transformation initiatives as a scientific consultant.

2019 – Present; **Research Assistant** – Department of Computer Science, University of the Western Cape, Cape Town, South Africa: **ACADEMIC INSTITUTION:**

- Responsible for supporting scientific research activities in the area of fourth industrial revolution within the Department of Computer Science under the supervision of Professor Isabela Venter.
- Assist in the utilisation of high performance computing related activities.

2019 – Present; **PhD Fellow**, University of the Western Cape, Cape Town, South Africa.

- Provide research investigation towards optimisation of deep learning algorithms for discrete irregular sequential analysis within the area of artificial intelligence.
- Design an implementable deep learning frameworks for irregular analysis as a solution to existing daily time series forecasting or prediction problems in complex scenarios. This is part of my PhD investigation.

2018 – 2020; **Director Projects'** – e-Government Technology Unit in the Office of the President and Cabinet, Government of the Republic of Zimbabwe, Harare, Zimbabwe: **PUBLIC SECTOR INSTITUTION:**

- Responsible for strategic planning, implementation and management of national, regional, continental and international digital transformation projects and programmes for an efficient Enterprise information architecture (EIA) ecosystem consisting of Unified Communication System (VoIP, DMS etc.), Cloud Services etc.
- In charge of managing G2C, G2B, G2G and G2E projects and programs i.e. technology stake, manufacturing, engineering and scientific projects with a cumulative value of more than US\$65 Million within set delivery timelines, costs, quality and scope.
- Formulating, mobilizing and managing Capital Expenditure (CAPEX) and Operational Expenditure (OPEX) finance and budgets of enterprise technology across Government.
- Responsible for managing over 50 e-Government stakeholders from Government Ministries, Departments and Agencies (MDAs) and Development Partners (World Bank,

IMF, United Nations, African Development Bank, ACBF) in line with national e-Government agenda based on establishment of digital and circular economy.

- In charge of the technical and operational management of digital technology landscape, infrastructure and platforms in line with set management protocols, Key Performance Indicators (KPIs) and Service Level Agreements (SLAs).
- Serving as the Technology Advisor for the whole Government digital transformation strategy, technology, services, delivery, operations and sustainability.
- Development and delivery of sustainable e-enable digital enterprise ecosystem projects, programs, policies and workflows using agile frameworks and methodologies that creates new ways of working (NWoW) across Ministries, Departments and Agencies (MDAs).
- Managing different stakeholder expectations through re-engineering, optimizing, simplifying and rationalising legacy processes and systems in an innovative way to improve execution services.

2014 – 2018; **Technical & Operations Manager** - Zimbabwe Centre for High Performance Computing (ZCHPC) at the University of Zimbabwe under the Ministry of Higher and Tertiary Education, Science and Technology Development of the Republic of Zimbabwe, Harare. Zimbabwe: **SCIENTIFIC RESEARCH—PUBLIC SECTOR INSTITUTION:**

- Serving as the technology officer by providing technical infrastructure and operational management of a tier 3 hybrid National High Performance Computing (HPC) Data Centre facility which include: maintenance of core cluster systems (computing, storage and networking i.e. server farms or nodes; firewalls, InfiniBand switches etc.) and auxiliary systems (electro-mechanical and security i.e. HVAC, UPS, Access Control, CCTV, Fire, standby Generators etc.)
- Managing Scientific Computing's computational and data science ecosystem of the HPC data centre facility by making sure the team of engineers, researchers and administrators monitor's infrastructure services, operations and performance in line with set procedures, protocols, standards and regulations
- Supervising 15 Data Centre technical and administration employees and over 300 Research Scientists and Engineers (Undergraduates to Post-Graduates) from Health Sciences, Engineering, Applied and Natural Sciences, Social Sciences, Arts, Business, Industry and Commerce in providing them with a state of the art high performance computing platform as a cloud service for advanced scientific research.
- Responsible for facility operational duties centered on day to day technical, engineering and management role of the High-Performance Computing facility. Overseeing data centre operational planning, cloud services, customer relations management (CRM), budgeting, procurement and maintenance of operational policies, procedures, standards, guidelines, best practices, contracts as well as Service Level Agreements (SLA's).
- Timely reporting to the Government through the Ministry of Higher and Tertiary Education, Science and Technology on the operational and utilisation status of the HPC facility through briefings, presentations, reports, workshops and seminars.
- Coordinating regional (SADC), Continental (Sub-Sahara) and Global High Performance Infrastructure projects and programs.

2010 – 2013; **Senior Research and Infrastructure Engineer** – Department of Science and Technology-Ministry of Higher and Tertiary Education, Science and Technology Development of the Republic of Zimbabwe, Harare, Zimbabwe, **PUBLIC SECTOR INSTITUTION:**

- Responsible for coordinating daily technical, research, innovation and development by providing IT technical and research support in form of installations, configurations, maintenance and upgrading of physical and virtual IT infrastructure environments composed of hardware, software, network and cybersecurity.
- Ensuring sound alignment of various ICT systems in line with public sector operations and demands by safeguarding effectiveness and sustainability of national cyber-infrastructure within the Department of Science and Technology.
- Responsible for establishment and supporting Zimbabwe Research & Education Network (ZimRen), ensuring network user growth planning, research software tools acquisition and development, asset management, information management, backup management, SLA's, contracts and IT policy and procedures.
- Supporting research, development and innovation projects, programs and initiatives within the context of science and technology.
- Playing a key role in advising architecture, designing and development of digital software and solutions as well as creation of enterprise services and applications.
- Pioneered and implemented the first Supercomputing, High-Performance Computing (HPC) Project in Zimbabwe.

### Research and Publications

#### Publications, Scientific Talks and Conference Presentations:

**K. Dandajena**, M. Ghaziasgar, I. M. Venter, and R. Dodds, (2022), "SeLFISA: A framework to enhance deep learning models for time series prediction," Submitted to the Scientific African, (2022), vol. 15, no. 7, Pages 1–20: Under Review

Batani J, Mbunge E, Muchemwa B, Gaobotse G, Gurajena C, Fashoto S, Kavu T, and **Dandajena K**, (2022), *A Review of Deep Learning Models for Detecting Cyberbullying on Social Media Networks*, 11<sup>th</sup> Computer Science On-line Conference 2022, April 2022

**Dandajena, K.**, Venter, I. M., Ghaziasgar, M. and R. Dodds (2021), *Selecting datasets for evaluating an enhanced deep learning framework*, SATNAC 21: Proceedings of the Annual Conference of the Southern African Telecommunication Networks and Applications Conference (SATNAC), Drakensberg, KwaZulu Natal, South Africa, November 2021.

**Dandajena. K**, Venter. I. M, Ghaziasgar M. and Dodds R. (2020), *Complex Sequential Data Analysis: A Systematic Literature Review*. Association for Computing Machinery (ACM).

**Dandajena. K**, Venter. I. M and Ghaziasgar M (2019), *An enhanced deep learning algorithm towards the accurate prediction of irregular sequential pattern*, Southern Africa Telecommunication Networks and Applications Conference (SATNAC) 2019, Durban, South Africa, September 2019 1-4.

**Dandajena K.**, (2016), *Research Growth through HPC in SADC Region*, SC 2016: The 28th Annual International Conference for High Performance Computing, Networking, Storage and Analysis, Salt Lake City, Utah, United States, November 13-18.

**Dandajena K.**, (2017), *SADC HPC Framework and Implementation Plan*, South African Development Community (SADC) HPC Collaboration Forum, 11<sup>th</sup> South African Centre for High Performance Computing (CHPC) Annual Conferences, Velmoré Hotel Estate, Pretoria, South Africa, Dec 4-6.

Ganda W.D & **Dandajena K.**, (2016), *Towards a Sustainable HPC System for Africa*, Southern African Development Community (SADC) HPC Forum, CHPC 10<sup>th</sup> International Conference, East London, Port Elizabeth, South Africa, December 1-5.

**Dandajena K.**, & Ganda W.D. (2017), *HPC Update on the Ranger Infrastructure Initiatives* -SADC Region, SC 2017: The 29<sup>th</sup> Annual International Conference for High Performance Computing or Supercomputing, Networking, Storage and Analysis, Denver, Colorado, United States, Nov12-17.

**Dandajena K.**, (2015), *Towards an integrated Cyber-infrastructure*, Southern African Development Community (SADC) HPC Forum, Center for High Performance Computing's (CHPC)- 10<sup>th</sup> National Meeting, CSIR ICC, Pretoria, South Africa, Nov30 – Dec 4.

**Dandajena K.**, (2015), *HPC Initiatives in Africa—Zimbabwe*, SC 2015: The 27<sup>th</sup> Annual Conference International Conference for High Performance Computing, Networking, Storage and Analysis, Austin, Texas, United States, November 15-20.

Ganda W. D, **Dandajena K.**, (2014) *Towards a Sustainable HPC System for Africa*, Southern African Development Community (SADC) HPC Forum, Center for High Performance Computing's (CHPC)- 9<sup>th</sup> National Meeting, Kruger National Park, Mpumalanga, South Africa, December 1-5.

**Dandajena K.**, Ganda W. D, (2013), *Human Capital Development for Developing Nations—Zimbabwe Case Study*, Southern African Development Community (SADC) HPC Forum, Center for High Performance Computing's (CHPC) - 8<sup>th</sup> National Meeting and Conference, Cape Town International Convention Centre, Cape Town, South Africa, December 2-6.

#### Research Focus:

My current research focus area is the optimization of deep learning algorithms in discrete irregular sequential patterns. Primary research interests lie in the areas of artificial intelligence, machine learning, deep learning, and related architecture, algorithms, and models through robust systematic frameworks. His area of research interest accelerates the development of converged, high speed, and miniaturized digital technological solutions that have a positive transformational impact on making the world a better place by resolving existing grand challenges.

#### Research Support and Funding:

2021 - Present; Telkom/Aria Technology Africa Centre of Excellence Research Support through the University of the Western Cape (ZAR24, 000. 00)/3 months.

2019 - 2021; Telkom/Aria Technology Africa Centre of Excellence Research Support through the University of the Western Cape (ZAR24, 000. 00)/3 months.

### **Teaching, Coaching, and Mentoring**

#### Course Director:

2020 – Present; COMS-RW-40315-1 (80 contact hours/ Semester): Artificial Intelligence. Two Cohorts per semester with 60 learners.

2020 – Present; COMS-RW-20315-1 (80 contact hours/semester): Discrete Mathematics. 2 Cohorts per semester with 60 learners.

#### Course Participation:

2020 – Present; COMS-RW-50920-1 (20 contact hours): Capstone Projects. Advising and mentoring 4<sup>th</sup> Year Computer Science Students on Capstone Research and Implementation Projects (10 Students/Semester) with specialty in Fourth Industrial Revolution and Digital transformation related projects. Most of these projects are in the domain of artificial intelligence, machine learning, digital transformation, high-end infrastructure and mathematics.

2020 – Present; Internal Course moderation for courses such as COMS-RW-20215-1-Mathematical Foundation of Computer Science (MFC), COMS-RW-50615-1-Data Mining, COMS-RW-50615-1-Machine Learning, COMS-RW-20815-1-Computer Architecture, COMS-RW-20515-1-Data Structures and Algorithms and COMS-RW-30115-1-Introduction to Systems Programming.

### Miscellaneous

#### Languages:

English: Fluent in writing, listening and speaking | Shona: Fluent in writing, listening and speaking | Zulu: Good in writing, listening and speaking | Ndebele: in writing, listening and speaking | French: Learning to write, listen and speak | Kinyarwanda: Basic in writing, listening and speaking

### Contact Information

**Employment Address :** African Institute for Mathematical Sciences (AIMS) | Global Secretariat

The Next Einstein Initiative | District Gasabo, Secteur Kacyiru,  
Cellule Kamatamu Rue KG590 ST. P.O Box: 6428,  
Kigali, Rwanda

**Postal Address:** African Institute for Mathematical Sciences (AIMS) | Global Secretariat

The Next Einstein Initiative | District Gasabo, Secteur Kacyiru,  
Cellule Kamatamu Rue KG590 ST. P.O Box: 6428,  
Kigali, Rwanda

**Residential Address:** KG 39 Avenue & KG158 Street,

Kimironko, Kigali  
Rwanda

Email: [kudadandaz@gmail.com](mailto:kudadandaz@gmail.com)

Skype: [kudadandaz1](https://www.skype.com/user/kudadandaz1)

Twitter : <https://twitter.com/kudadandaz>

LinkedIn: <https://www.linkedin.com/in/kudakwashe-dandajena-b2510860>

GitHub: <https://github.com/Dandajena>

WhatsApp: +250785821565



## References

Fay Hodza, PhD. **Executive Director**, Global Teachers Institute, | Email [fhodza@globalteachers.org](mailto:fhodza@globalteachers.org) | Tel +27 (0)11 403 6401 | Cell +27 731776639 (Former Academic Director-African leadership University)

Isabella M. Venter, PhD: **Professor Emeritus-Academic Supervisor**, Department of Computer Science, University of the Western Cape, South Africa | Email: [venter@uwc.ac.za](mailto:venter@uwc.ac.za) | Cell: +27 82 202 3520

Willie Ganda, PhD: **Chief Director**, Ministry of Higher & Education, Science and Technology Development, Zimbabwe. | E-mail: [gandawd@gmail.com](mailto:gandawd@gmail.com) | Cell: +263 716 801 303.